

AI For Math Seminar, ZJU

Lec 06: Formalize and Verify Algorithm and Theorem

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Outline

- 1 Introduction
 - Methodology
- 2 Termination Proofs
- 3 Algorithm Complexity and CSLib

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- Similarly, in software engineering, we can write an infinite number of functions that receive the same type of input and produce some output. How can we rigorously know which one is correct?
- In this section, we first recall foundational knowledge in Computer Science, then we translate these concepts into Lean 4, learning the practical skills to verify real projects.

Correctness and Termination of an Algorithm 1

The fundamental question: What does it mean for an algorithm to be “correct”?

- Under the Curry-Howard Isomorphism, we already know that an algorithm corresponds to the construction of a mathematical theorem. So the question becomes: what is the correctness of a theorem?

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- In the List Sorting problem, we want a procedure to receive an arbitrary-length finite list and return a sorted list. Here, “correctness” is formally defined as maintaining the permutation of elements while satisfying a mathematical ordering property.
- Therefore, we need to define correctness manually. In Lean, we encode the human intent of “correctness” as a mathematically rigorous **Proposition**.

Correctness and Termination of an Algorithm 2

Let's review the classical CS definitions and prepare to map them to Lean.

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- **Total Correctness:** For any given specification, we say an algorithm is **totally correct** if and only if:
 - 1 It always terminates (Halting).
 - 2 It returns the correct result according to the specification.
- As a result, to formally verify an algorithm, our proof burden is twofold: we must prove **Termination** and **Semantic Correctness**.

Exercise in Lean: Formulating the Problem

In Lean, every function is inherently required to be totally correct. You simply cannot compile a function that might infinitely loop without an escape hatch. Let's tackle Correctness first.

- **Translating the Specification:** Because Lean allows us to use Propositional Logic as a type (Dependent Type Theory), we declare “correctness” through a Theorem signature. What we must check is whether our Theorem actually captures the meaning we want.

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- **Exercise:** Write a recursive function to sum up an arbitrary `List Nat`. (We ignore hardware overflow limits here; Lean's `Nat` is arbitrarily large).

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- **Exercise:** Write a recursive function to sum up an arbitrary `List Nat`. (We ignore hardware overflow limits here; Lean's `Nat` is arbitrarily large).
- Once written, how do we mathematically prove it behaves like a sum?

Exercise in Lean: Writing the Code

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First, we define the computational procedure. We use structural recursion over the list.

```
def listSum (xs : List Nat) : Nat :=  
  match xs with  
  | [] => 0  
  | y :: ys => y + listSum ys
```

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```

Next, we must invent a **Specification** (Theorem). A good way to prove a sum is correct is to prove it distributes over list concatenation (`++`):

```
theorem listSum_append (xs ys : List Nat) :  
  listSum (xs ++ ys) = listSum xs + listSum ys := by  
  -- Proof goes here
```

Solution Strategy: Induction

Our Task: Prove the correctness of the hypothesis given our defined type.

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- In more complex scenarios, our strategy is to use the `cases` or `match` tactics to logically divide the type into exhaustive branches, and finish the proof for each branch.
- For example, using a theorem like “Every natural number is either even or odd” to introduce new hypotheses, allowing us to traverse deeper into number theory proofs.

Solution Execution: The Proof

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We apply structural induction on the first list, `xs`.

```

theorem listSum_append (xs ys : List Nat) :
  listSum (xs ++ ys) = listSum xs + listSum ys := by
  induction xs with
  | nil =>
    -- Base case: listSum ([]) ++ ys) = listSum [] + listSum ys
    -- This is trivially true by definition of listSum.
    rfl
  | cons head tail ih =>
    -- Inductive step: we assume the theorem holds for `tail` (ih).
    calc
      listSum ((head :: tail) ++ ys)
        = head + listSum (tail ++ ys)           := by rfl
      _ = head + (listSum tail + listSum ys) := by rw [ih]
      _ = (head + listSum tail) + listSum ys := by omega
      _ = listSum (head :: tail) + listSum ys := by rfl

```

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- The final goal of formal verification is to handle cases where algorithms do *not* trivially shrink.

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- Why didn't Lean ask us for a termination proof for `listSum`? Because Lean automatically detected that `tail` is strictly smaller than `head :: tail` (Structural Recursion).
- The final goal of formal verification is to handle cases where algorithms do *not* trivially shrink.
- In Lean, proving termination is equivalent to writing a well-defined function that passes the kernel's strictly descending metrics without a compiler error. This brings us to Section 2.

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- British mathematician, the father of theoretical computer science
- **The Halting Problem** (1936): no general algorithm can decide whether an arbitrary program will eventually halt or loop forever
- Proof by contradiction using a self-referential program — the original undecidability result
- **Consequence for us:** termination cannot be automated for all programs in a Turing-complete language. Lean instead *requires* you to prove it.

The Philosophy of Totality

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- The Halting Problem states that no compiler can statically guarantee termination for all programs in a Turing-complete language.
- **The Lean 4 View:** Lean strictly requires *Totality*. Every defined function **must** terminate in a finite number of steps for all possible inputs.
- **Why?** Because of the Curry-Howard Isomorphism.

Why Infinite Loops Destroy Mathematics

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- If you can prove `False`, you can use the Principle of Explosion (*ex falso quodlibet*) to prove $1 = 0$, $P \wedge \sim P$, or any other absurdity.
- **Conclusion:** Proving termination is not just a performance optimization; it is a fundamental requirement for mathematical soundness.

The Happy Path: Structural Recursion

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For most data structures, Lean proves termination automatically by observing **Structural Decreasing**.

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- Inductive types in Lean are strictly finite trees. You cannot descend into a finite tree infinitely.
- Lean's internal heuristics accept this without any manual proof.

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What happens when an algorithm shrinks its input *mathematically*, rather than *structurally*?

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def log2 (n : Nat) : Nat :=
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unproved application: log2 (n / 2)

Lean sees $n / 2$. To the compiler, this is an arbitrary function call. It does not natively know that division by 2 produces a smaller integer.

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- **Mathematical Definition:** A relation $<$ on a set S is well-founded if every non-empty subset of S has a minimal element with respect to $<$.
- **Equivalently:** There are no countably infinite descending chains $x_0 > x_1 > x_2 > \dots$
- **The Gold Standard:** The strictly-less-than relation ($<$) on Natural Numbers ($\mathbb{N}$). Because you cannot go below 0, any sequence of strictly decreasing natural numbers must eventually stop.

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We guide Lean by defining a **Measure** — a mapping from the function's arguments to a `Nat` that strictly decreases.

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- Notice the `h : n < 2`. We bind the condition to a hypothesis!
- In the `else` branch, Lean's context now contains `Not (n < 2)`, meaning $n \geq 2$. Lean's automated arithmetic can now prove $(n/2) < n$.

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def slow_div (n k : Nat) : Nat :=  
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Lean cannot prove this blindly. If $k = 0$, then $n - 0$ is not less than n . We must intervene manually.

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The Power of Context: Because we saved the branch conditions as `h0` and `h1`, `omega` sees that $k \geq 1$ and effortlessly proves $n - k < n$.

Live Lean: `D6_Termination.lean`

Advanced Measures: Beyond Arguments

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The measure doesn't have to be a direct parameter. It can be a computed expression representing the "remaining work".

```
-- Traversing an array from index `i` to the end
def process {A : Type} (arr : Array A) (i : Nat) : Nat :=
  if h : i < arr.size then
    1 + process arr (i + 1)
  else
    0
-- The parameter `i` is INCREASING, so it can't be the measure!
termination_by arr.size - i
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Insight: We map the state to a strictly decreasing distance:
`arr.size - i`.

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What happens when algorithms decrease alternately across multiple parameters? The classic example is the **Ackermann function**.

```
def ack (m n : Nat) : Nat :=  
  match m, n with  
  | 0, n => n + 1  
  | m' + 1, 0 => ack m' 1  
  | m' + 1, n' + 1 => ack m' (ack (m' + 1) n')
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Notice the chaotic parameter flow:

- Sometimes m goes down, and n resets.
- Sometimes m stays the same, and n goes down.
- Sometimes n turns into a massive recursive call!

Lexicographic Ordering (Dictionary Order)

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- Think of alphabetical sorting: "apple" comes before "apply". You check the first letter. If it ties, you check the second.

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- To prove ack terminates, Lean uses a **Lexicographic Order** across a tuple of variables.
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- **Math Definition for pairs (m, n) :**
 $(m_2, n_2) < (m_1, n_1)$ if and only if:
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$$m_2 < m_1 \quad \text{OR} \quad (m_2 = m_1 \quad \text{AND} \quad n_2 < n_1)$$
- As long as the tuple strictly decreases under this definition, infinite loops are mathematically impossible.

Syntax for Lexicographic Measures

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To trigger Lexicographic ordering, we pass multiple identifiers to `termination_by`.

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def ack (m n : Nat) : Nat :=  
  match m, n with  
  | 0, n => n + 1  
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termination_by m n
```

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- Only if m is exactly equal, will Lean look at `n`.

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Let's trace the hardest branch: $| m' + 1, n' + 1 \Rightarrow \text{ack } m'$
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- **Lean stops checking.**
- It completely ignores parameter 2. This is brilliant because parameter 2 is `ack (m' + 1) n'`, which is a massive number that Lean cannot statically evaluate!

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This fails completely.

- For the outer call, Lean checks n first.
- Old n was $n' + 1$. New n is $\text{ack_broken } (m' + 1) n'$.
- Lean cannot prove that the output of Ackermann is smaller than $n' + 1$. The termination proof crashes.

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- Lean supports **Mutual Recursion** using the `mutual` keyword.
- The termination proof requires finding a unified measure that decreases across *both* functions.
- *(For the sake of time in this seminar, we omit the deep dive into mutual termination, but it relies on exactly the same `termination_by` principles mapped to a sum type).*

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- 5 When automated arithmetic fails, open a `decreasing_by` block and unleash `omega`.

Outline

- 1 Introduction
 - Methodology
- 2 Termination Proofs
- 3 Algorithm Complexity and CSLib

Part 3: Algorithm Complexity and CSLib

Part 3: Algorithm Complexity & CSLib

Cost Models, Asymptotic Big-O, and Automated Complexity Proofs

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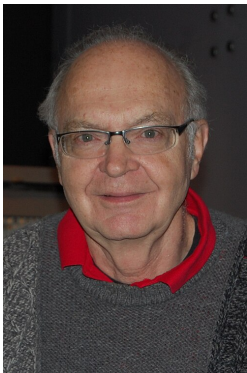
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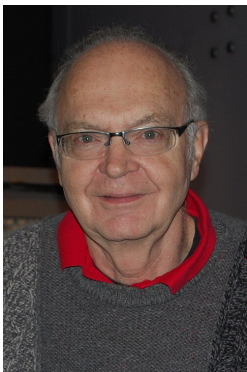
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- In mission-critical environments (e.g., Blockchain Smart Contracts, Real-time OS like seL4), a function that takes $O(2^n)$ time to terminate is functionally equivalent to an infinite loop. It is a Denial of Service (DoS) vulnerability.
- Therefore, we must formally prove **Time and Space Complexity** as a mathematical theorem.

Donald Knuth (1938–) and Algorithm Analysis

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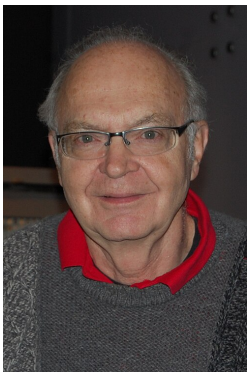


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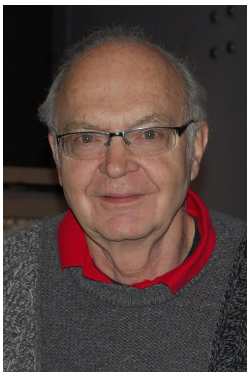
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- In this seminar, we focus on the **Extrinsic** method, as it separates the complexity proof from the correctness proof.

Extrinsic Cost Modeling: Binary Search

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Let's model the worst-case step count of **binary search** on a sorted array.

-- The pure algorithm: $O(\log n)$ comparisons

```
def bsearch (arr : Array Nat) (t : Nat) (lo hi : Nat) : Option Nat :=
  if lo > hi then none
  else
    let mid := lo + (hi - lo) / 2
    if arr[mid]! == t then some mid
    else if arr[mid]! < t then bsearch arr t (mid + 1) hi
    else bsearch arr t lo (mid - 1)
termination_by hi - lo + 1
```

-- Parallel step counter: tracks recursive depth only

```
def time_bs : Nat -> Nat -- argument = interval width n = hi - lo +
  | 0 => 1 -- empty interval: one step (check lo > hi)
  | n + 1 => 1 + time_bs ((n + 1) / 2) -- one step + recurse on half
termination_by n
```

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- **Notation:** `f =O[atTop] g`

Stating the Complexity Theorem

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We unfold the recurrence $T(n) = T(\lfloor n/2 \rfloor) + 1$ to get
 $T(n) = \lfloor \log_2 n \rfloor + 1$.

-- 1. Closed-form bound: log n + 1 steps in the worst case

```
def T_bs (n : Nat) : Nat := Nat.log 2 n + 1
```

-- 2. State the Big-O theorem

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open Asymptotics Filter
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theorem bsearch_is_O_log :
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Key point: `Nat.log 2 n` in Lean is the floor of $\log_2 n$ — the number of times you can halve n before reaching 0.

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- This beats $O(n^2)$ for large n — and the exponent is *not an integer*!

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While proving $O(\log n)$ is manageable, recurrences like Karatsuba's are a nightmare manually.

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 - Sum the geometric series $\sum_{k=0}^{\log_2 n} 3^k (n/2^k)$ and apply $n^{\log_2 3}$ bound
- Hundreds of lines of Lean for a single complexity theorem.
- If you tweak the constant factor, the whole proof must be rederived.

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While proving $O(\log n)$ is manageable, recurrences like Karatsuba's are a nightmare manually.

- To prove $T(n) = 3T(n/2) + cn \implies T(n) = O(n^{\log_2 3})$ in Lean, you must:
 - Handle $\lfloor n/2 \rfloor$ vs $\lceil n/2 \rceil$ for odd n
 - Unwind the recurrence to depth $\log_2 n$, accumulate 3^k terms per level
 - Sum the geometric series $\sum_{k=0}^{\log_2 n} 3^k (n/2^k)$ and apply $n^{\log_2 3}$ bound
- Hundreds of lines of Lean for a single complexity theorem.
- If you tweak the constant factor, the whole proof must be rederived.
- **We need specialized infrastructure.** Enter **CSLib**.

Introducing CSLib: The Missing Piece

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 - **Imperative Verification:** Separation Logic frameworks for verifying C-like code with pointers and memory mutations.

The Frontier: Automated Complexity Proofs

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Modern verification (like the tools bundled in or inspired by CSLib) aims to completely **automate** the extraction and proof of recurrences.

- **Step 1: Syntactic Extraction.** Using Lean 4's powerful macro system, we can write a metaprogram that reads the AST of your `def` and automatically generates the `time_` function.

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- **Step 2: Recurrence Solving.** Instead of manual induction, the library provides a formalized **Master Theorem** (or Akra-Bazzi method).
- **Step 3: Tactic Integration.** A custom tactic (e.g., `complexity`) matches the extracted recurrence against the Master Theorem database and discharges the proof instantly.

What Automated Complexity Looks Like (Conceptual)

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With modern CS libraries, the boilerplate vanishes. The user experience approaches the simplicity of writing a type signature.

```
-- Annotate Karatsuba so the framework auto-generates the cost model
@[derive_cost]
def karatsuba (x y : BigNat) : BigNat :=
  ... -- 3 recursive calls on n/2-digit halves

-- State the theorem with a single auto-tactic
-- The framework extracts  $T(n) = 3T(n/2) + cn$ , applies
-- the formalized Master Theorem, and closes the goal.
theorem karatsuba_complexity :
  cost_karatsuba = 0[atTop] (fun n => n ^ Real.log 3 / Real.log 2) := b
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Note: This is the active research frontier. Tools like AutoBound and Wormhole are actively bridging the gap between theoretical CS and Lean's automation.

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- ➍ **CSLib Ecosystem:** Provides the required theorems (like the Master Theorem) and verified data structures that pure Mathlib lacks.
- ➎ **Automation is Key:** Metaprogramming in Lean 4 allows us to extract cost recurrences and solve them automatically, hiding the tedious math from the software engineer.

Final Project Framework

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Designing an Automated Complexity Prover using the TimeM Monad

Project Motivation: Intrinsic vs. Extrinsic

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- **The Solution:** The **Intrinsic** approach. We embed step-counting directly into the execution context of the function using a custom Monad.
- **Your Final Project:** You will implement a step-counting monad called `TimeM`, write standard algorithms within it, and build a verification pipeline to automate their asymptotic complexity proofs.

Defining the TimeM Monad

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Under the hood, TimeM is simply a state monad that carries an accumulating natural number representing "ticks" or execution cost.

```
-- Define the stateful computation type
def TimeM (A : Type) : Type := Nat -> A * Nat

-- Making it a Monad so we can use do-notation
instance : Monad TimeM where
  pure x := fun c => (x, c)
  bind x f := fun c =>
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We define a foundational primitive operation to increment our clock:

```
def tick : TimeM Unit :=
  fun c => ((), c + 1)
```


Writing Algorithms inside TimeM

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Using Lean's do-notation, we can write algorithms that look identical to standard imperative code, explicitly instrumenting our cost metrics.

```
def insertM (x : Nat) (xs : List Nat) : TimeM (List Nat) :=
  match xs with
  | [] => return [x]
  | y :: ys => do
    tick -- Charge 1 cost for the comparison operation
    if x <= y then
      return x :: y :: ys
    else do
      let rest <- insertM x ys
      return y :: rest
```

Extracting the Cost Function

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To verify the execution cost, we need to evaluate the monad starting from an initial clock cycle of 0, discarding the computed output value.

```
-- Helper to extract ONLY the accumulated cost
def getCost {A : Type} (comp : TimeM A) : Nat :=
  (comp 0).2

-- Example evaluation
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The Mathematical Target: For any list `xs`, we want to formally prove an upper bound on `getCost (insertM x xs)` proportional to the list's length.

Live Lean: `D6_TimeM.lean`

Project Milestone 1: Manual Monadic Bounds

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Your first milestone is to manually prove that the monadic implementation satisfies a precise mathematical bound before mapping it to Big-O.

```
theorem insertM_cost_bound (x : Nat) (xs : List Nat) :  
  getCost (insertM x xs) <= xs.length := by  
  induction xs with  
  | nil =>  
    -- Base case: length is 0, cost is 0  
    rfl  
  | cons y ys ih =>  
    -- Inductive step: unfolds the match and uses  
    -- the inductive hypothesis (ih)  
    unfold insertM getCost  
    ...
```

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- **Goal:** Create a macro or a unified helper tactic that can instantly discharge basic complexity bounds for structural steps.
- **Mechanism:** Monadic code written with simple structural recursion creates highly predictable cost equations during unfolding.
- You will compile a hint database of lemmas showing how `tick` interacts with `bind`, allowing Lean's simplifier (`simp`) to flatten monadic state tracking automatically.

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By combining structural induction with a customized rewriting sequence, we can create a composite tactic sequence that solves these goals seamlessly.

```
-- Conceptual pipeline for your custom automation
macro "prove_complexity" : tactic => `(tactic|
  intros;
  macro_rules | `(tactic| leaves) => `(tactic| rfl)
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```

Your project must demonstrate that this automated tactic can successfully close the complexity theorems for at least three distinct algorithms: Linear Search, Insertion Sort, and Binary Tree lookup.

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- 3 **The Automation Engine:** A custom Lean 4 macro or proof script that handles structural cost simplification without manual intervention per branch.
- 4 **Asymptotic Integration:** Final proofs framing your extracted monadic costs within Mathlib's asymptotic Big-O filter definitions (`=0[atTop]`).